

Displays employees who are not IT programmers and whose salary is less than that of any IT programmer. The maximum salary that a programmer earns is \$9,000.

< ANY means less than the maximum. > ANY means more than the minimum. = ANY is equivalent to IN.

Using the ALL Operator in Multiple-Row Subqueries

```
SELECT employee_id, last_name, job_id, salary
FROM employees
WHERE salary < ALL (SELECT salary FROM employees WHERE job_id = 'IT_PROG')
AND job_id <> 'IT_PROG';
```

Displays employees whose salary is less than the salary of all employees with a job ID of IT_PROG and whose job is not IT_PROG.

> ALL means more than the maximum, and < ALL means less than the minimum.

The NOT operator can be used with IN, ANY, and ALL operators.

Null Values in a Subquery

```
SELECT emp.last_name FROM employees emp
WHERE emp.employee_id NOT IN (SELECT mgr.manager_id FROM employees mgr);
```

Notice that the null value as part of the results set of a subquery is not a problem if you use the IN operator. The IN operator is equivalent to = ANY. For example, to display the employees who have subordinates, use the following SQL statement:

```
SELECT emp.last_name
FROM employees emp
WHERE emp.employee_id IN (SELECT mgr.manager_id FROM employees mgr);
```

Display all employees who do not have any subordinates:

```
SELECT last_name FROM employees
WHERE employee_id NOT IN (SELECT manager_id FROM employees WHERE manager_id IS
NOT NULL);
```

Find the Solution for the following:

1. The HR department needs a query that prompts the user for an employee last name. The query then displays the last name and hire date of any employee in the same department as the employee whose name they supply (excluding that employee). For example, if the user enters Zlotkey, find all employees who work with Zlotkey (excluding Zlotkey).

Select (-have, hire-date) from emp where dept-id = (select dept-id from emp where (-have - 'Zlotkey')) and (-have - 'Zlotkey');

2. Create a report that displays the employee number, last name, and salary of all employees who earn more than the average salary. Sort the results in order of ascending salary.

Select emp-id, (-have, salary) from employees where salary > (select avg(salary) from emp) order by salary asc;

3. Write a query that displays the employee number and last name of all employees who work in a department with any employee whose last name contains a u.

```
select emp-id, last-name from emp1 where dept-id in  
(select dept-id from emp1 where (last-name like '%u%'));
```

4. The HR department needs a report that displays the last name, department number, and job ID of all employees whose department location ID is 1700.

```
select last-name, dept-id, job-id from emp1 where job-id is  
(select job-id from emp1 where department-id in (select dept-id  
from dept where (loc-id=1700)));
```

5. Create a report for HR that displays the last name and salary of every employee who reports to King.

```
select last-name, salary from emp1 where manager-id =  
(select emp-id from emp1 where last-name = 'king');
```

6. Create a report for HR that displays the department number, last name, and job ID for every employee in the Executive department.

```
select dept-id last-name, job-id from emp1 where dept-id  
= (select dept-id from dept where dept-name = "Executive");
```

7. Modify the query 3 to display the employee number, last name, and salary of all employees who earn more than the average salary and who work in a department with any employee whose last name contains a u.

```
select emp-id, last-name, sal from emp1 where sal >  
(select avg (salary) from emp1 and dept-id in (select  
distinct dept-id from emp1 where (last-name like '%u%')));
```


Practice Questions

1. Ellen Abel is an employee who has received a \$2,000 raise. Display her first name and last name, her current salary, and her new salary. Display both salaries with a \$ and two decimal places. Label her new salary column AS New Salary.

```
SELECT first_name, last_name, TO_CHAR(salary, 'Salary,') AS Current Salary,
TO_CHAR(salary + 2000, 'Salary,') AS "New Salary" FROM employees
WHERE first_name = 'Ellen' AND last_name = 'Abel';
```

2. On what day of the week and date did Global Fast Foods' promotional code 110 Valentine's Special begin?

```
SELECT TO_CHAR(Start_date, 'Day, Month DD, YYYY') AS "Promo
Start Date" FROM d_promos WHERE Promo_Code = 110;
```

3. Create one query that will convert 25-Dec-2004 into each of the following (you will have to convert 25-Dec-2004 to a date and then to character data):

December 25th, 2004

DECEMBER 25TH, 2004

25th december, 2004

SELECT

```
TO_CHAR(TO_DATE('25-DEC-2004', 'DD-MON-YYYY'),
'FM MONTH DDTH, YYYY') AS "Format 1",
TO_CHAR(TO_DATE('25-DEC-2004', 'DD-MON-YYYY'),
'FM MONTH DDTH, YYYY') AS "Format 2",
TO_CHAR(TO_DATE('25-DEC-2004', 'DD-MON-YYYY'),
'FM MONTH Month, YYYY') AS "Format 3" FROM dual;
```

4. Create a query that will format the DJs on Demand d_packages columns, low-range and high-range package costs, in the format \$2500.00.

SELECT

```
code,
TO_CHAR(low_range, '$99,999.00') AS "Low Range Cost",
TO_CHAR(high_range, '$99,999.00') AS "High Range Cost"
```

5. Convert JUNE192004 to a date using the fx format model. FROM d_packages;

SELECT

```
TO_DATE('JUNE192004', 'FXMONTHDDYYYY') AS "Converted Date"
FROM
dual;
```

6. What is the distinction between implicit and explicit datatype conversion? Give an example of each.

SAL: SELECT implicit: SELECT * FROM employees WHERE employee_id = '101';

explicit: SELECT * FROM employees WHERE hire_date > TO_DATE('01-JAN-2023', 'DD-MON-YYYY');

7. Why is it important from a business perspective to have datatype conversions?

* Data Integration (ETL) = Businesses collect data from many sources which arrives in the correct format. ETL is necessary to extract, Transform and Load this data

User friendly reporting: used to present data in a well easy to read manner ex:- 2025 10 29 as "October 29 2025")

Accurate calculations &

✓ Comparisons:- To perform business calculation data types must match ensures to calculate totals, averages and time based metrics.

Evaluation Procedure	Marks awarded
Practice Evaluation (5)	5
Viva(5)	5
Total (10)	10
Faculty Signature	